

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I S. Santhiya (192511360) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Santhiya

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately 2	1.5 Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

	Problem understanding (20%)	method selection (20%)	Solution Process (20%)	Accuracy of calculation (20%)	Interpretation of result (15%)	
Q ₁	2	1.5	2	1.5	1.5	8.5
Q ₂	1.5	1	1.5	1.5	1	6.5
Q ₃	2	2	2	2	1	9
Q ₄	1	1	1.5	1	1	5.5
Q ₅	2	1	1	1	1.5	6.5

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1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answers:

Subnetting Calculation:

Given

Network: 192.168.20.0/24

- Lab A → /26
- Lab B → /27
- Lab C → /28

Step 1: Lab A — /26

Subnet Mask: /26 = 255.255.255.192

Host bits: $32 - 26 = 6$

Total addresses: $2^6 = 64$

Usable hosts: $64 - 2 = 62$

Block size: $256 - 192 = 64$

Therefore:

- Network Address = 192.168.20.0
- First Host = 192.168.20.1

- Last Host = 192.168.20.62
- Broadcast = 192.168.20.63

Step 2: Lab B — /27

Next available network = 192.168.20.64

Subnet Mask: /27 = 255.255.255.224

Host bits: $32 - 27 = 5$

Total addresses: $2^5 = 32$

Usable hosts: $32 - 2 = 30$

Block size: $256 - 224 = 32$

Therefore:

- Network Address = 192.168.20.64
- First Host = 192.168.20.65
- Last Host = 192.168.20.94
- Broadcast = 192.168.20.95

Step 3: Lab C — /28

Next available network = 192.168.20.96

Subnet Mask: /28 = 255.255.255.240

Host bits: $32 - 28 = 4$

Total addresses: $2^4 = 16$

Usable hosts: $16 - 2 = 14$

Block size: $256 - 240 = 16$

Therefore:

- Network Address = 192.168.20.96
- First Host = 192.168.20.97
- Last Host = 192.168.20.110
- Broadcast = 192.168.20.111

Lab	CIDR	Network address	Host range	Broad cast	Usable host
A	/26	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
B	/27	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
C	/28	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Total usable hosts = 62 + 30 + 14 = 106

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answers:

Network Administrator

Step-by-Step Solution

Given

- System 1 IP: 192.168.1.130
- System 2 IP: 192.168.1.190
- Subnet Mask: 255.255.255.192
- CIDR: /26

Step 1: Block Size

$$256 - 192 = 64$$

Subnet ranges:

- 192.168.1.0 – 63
- 192.168.1.64 – 127
- 192.168.1.128 – 191
- 192.168.1.192 – 255

Step 2: System 1

IP = 192.168.1.130

- Network Address: 192.168.1.128
- Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191
- Usable Hosts: 62
- IP is valid: Yes

Step 3: System 2

IP = 192.168.1.190

- Network Address: 192.168.1.128
- Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191
- Usable Hosts: 62
- IP is valid: Yes

System	Network	Host Range	Broad Cast	Usable Host
1	192.168.1.128	.129 – .190	.191	62
2	192.168.1.128	.129 – .190	.191	62

Both IP addresses are in the same subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Answers:

Given

Network: 172.16.0.0/16

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Step 1: HR – /25

/25 = 255.255.255.128

Block size:

$$256 - 128 = 128$$

- Network Address: 172.16.0.0
- Host Range: 172.16.0.1 – 172.16.0.126
- Broadcast: 172.16.0.127
- Usable Hosts: 126

Step 2: Sales – /26

Next available address = 172.16.0.128

/26 = 255.255.255.192

Block size:

$$256 - 192 = 64$$

- Network Address: 172.16.0.128
- Host Range: 172.16.0.129 – 172.16.0.190
- Broadcast: 172.16.0.191
- Usable Hosts: 62

Step 3: IT – /27

Next available address = 172.16.0.192

/27 = 255.255.255.224

Block size:

$$256 - 224 = 32$$

- Network Address: 172.16.0.192
- Host Range: 172.16.0.193 – 172.16.0.222
- Broadcast: 172.16.0.223
- Usable Hosts: 30

Step 4: Admin – /28

Next available address = 172.16.0.224

/28 = 255.255.255.240

Block size:

$$256 - 240 = 16$$

- Network Address: 172.16.0.224
- Host Range: 172.16.0.225 – 172.16.0.238
- Broadcast: 172.16.0.239
- Usable Hosts: 14

Department	Subnet	Network Address	Host Range	Broad cast	Usable Host
HR	/25	172.16.0.0	1-126	127	126
Sales	/26	172.16.0.128	129-190	191	62
IT	/27	172.16.0.192	193-222	223	30
Admin	/28	172.16.0.224	225-238	239	14

Remaining Unused IP Range

The allocated network is: 172.16.0.0 – 172.16.255.255

The departments use: 172.16.0.0 – 172.16.0.239

Therefore, the remaining unused range is: 172.16.0.240 – 172.16.255.255

Total usable hosts allocated:

$$126 + 62 + 30 + 14 = 232$$

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Answers:

Given

- Network: 172.16.10.0/24
- New subnet mask: /27
- IP address: 172.16.10.78
- Check IP: 172.16.10.95

Step 1: Number of Subnets

Original prefix = /24

New prefix = /27

Borrowed bits: $27 - 24 = 3$

Number of subnets: $2^3 = 8$ subnets

Step 2: Usable Hosts per Subnet

Host bits: $32 - 27 = 5$

Total addresses: $2^5 = 32$

Usable hosts: $32 - 2 = 30$ hosts

Step 3: Find the Subnet of 172.16.10.78

/27 = 255.255.255.224

Block size:

$$256 - 224 = 32$$

Subnet ranges:

- 0 – 31
- 32 – 63
- 64 – 95
- 96 – 127
- 128 – 159
- 160 – 191
- 192 – 223
- 224 – 255

78 falls in 64–95.

Therefore:

- Network Address: 172.16.10.64
- Host Range: 172.16.10.65 – 172.16.10.94
- Broadcast Address: 172.16.10.95

Step 4: Check 172.16.10.95

- 172.16.10.95 is the broadcast address.
- Therefore, it belongs to the same subnet range, but it cannot be used as a host IP.

Item	Answer
Number of subnets	8
Usable Hosts/Subnet	30
IP Address	172.16.10.78
Network Address	172.16.10.64
Host Range	172.16.10.65-172.16.10.94
Broad Cast	172.16.10.95
172.16.10.95 same subnet?	Yes but it is the broadcast address

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answers:

Given IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix length = /64

Step 1: Understand IPv6

An IPv6 address has 8 groups, and each group has 4 hexadecimal digits.

2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 : 8329

Step 2: Compress the Address

We can remove leading zeros from each group:

2001:db8:0:0:0:ff00:42:8329

There are three continuous 0 groups:

0:0:0

These can be replaced by ::.

So, the shortest form is:

2001:db8::ff00:42:8329

Step 3: Expand It Again

The :: represents the missing zero groups.

So:

2001:db8::ff00:42:8329

becomes:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 4: Find the Network Prefix

/64 means the first 64 bits are used for the network.

Since each IPv6 group has 16 bits:

$$16 \times 4 = 64 \text{ bits}$$

So the first 4 groups are the network:

2001:0db8:0000:0000

Therefore:

Network Prefix = 2001:db8::/64

Step 5: Find Host Addresses

IPv6 has 128 bits.

Network bits = 64

Host bits:

$$128 - 64 = 64$$

Number of possible host addresses:

$$2^{64} = 18,446,744,073,709,551,616$$

Final Answer

- Compressed: 2001:db8::ff00:42:8329
- Expanded: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Network Prefix: 2001:db8::/64
- Host Bits: 64
- Possible Host Addresses: 18,446,744,073,709,551,616

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